MIL-A-22681C(SHIPS)
SUPPLEMENT-1A
24 March 1972
SUPERSEDING
SUPPLEMENT-1
15 September 1970

MILITARY SPECIFICATION

ARRESTER, FLASH, MAIN STORAGE BATTERY,

(SUBMARINE USE), GENERAL SPECIFICATION FOR

This supplement lists the Specification Sheets associated with Military Specification MIL-A-22681C (SHIPS), 24 March 1972.

Specification sheets	Title
MIL-A-22681/1C(SHIPS)	Arrester, Flash, Submarine Main Storage Battery, Type I.
MIL-A-22681/2C(SHIPS)	Arrester, Flash, Submarine Main Storage Battery, Type II.
MIL-A-22681/3B(SHIPS)	Arrester, Flash, Submarine Main Storage Battery, Type III.

Preparing activity:
Navy - SH

FSC 6140

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Table I - Maximum impurities permitted in refluxed solution (cont'd).

Material	Weight percent
Iron	.089
Manganese	.0004
Nitrates (as NO_3)	.0146
Platinum	None
Selenium	None
Sulfites (as SO_2)	None
Zinc	None
Ammonium (NH_4)	(Report)
Calcium and magnesium oxides	0.022
Organic and volatile material	.029
Suspended matter	Lead compounds only

3.10 Silicone resin cure. Silcone resin cure shall be determined by the hot acid test specified in 4.5.7.

3.11 Dimensions. All dimensions for the flash arrester and component parts shall be as specified on the applicable specification sheet.

3.12 Weight. Flash arresters complete with component parts shall not exceed the weight specified on the applicable specification sheet.

3.13 Marking. Flash arresters shall be marked in accordance with MIL-STD-130 and include the manufacturer's name and the month and year of manufacture.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The supplier shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.1 and 6.2).

4.2 Qualification tests.[1/] Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center (NAVSEC). Qualification tests shall consist of the examination and tests specified in 4.4 and 4.5.1 through 4.5.7.

4.2.1 Qualification samples. Three sample flash arresters of any type for which qualification is desired shall be submitted. Qualification approval of the type submitted will include qualification of all types. Varying the seal material or method of preparing surfaces for sealing shall be considered to result in a new construction of flash arrester.

4.2.2 Qualification sample data. Each manufacturer submitting qualification samples shall also submit a complete description of his product, which shall include the following:

(a) Names of manufacturers of component parts.
(b) Complete description of sealing compounds, primers and catalysts used to make the seal between dome and base.
(c) Certification that the manufacturer, if his product qualifies, will supply on procurements, flash arresters manufactured with the same materials and by the same methods as those used for the qualification samples.

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.3.1).

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4.3 Quality conformance inspection.

4.3.1 Lot. A lot shall consist of 600 flash arresters.

4.3.2 Sampling for examination. Samples for examination (see 4.4), shall be selected in accordance with MIL-STD-105 at inspection level II. The acceptable quality level (AQL) shall be 2.5 percent.

4.3.3 Sampling for tests. Four samples from each lot shall be selected at random and subjected to the tests of 4.5.4 through 4.5.7. The complete impurities test of 4.5.6 shall be made on 3 additional samples selected from the first lot of each contract or order. Samples from subsequent lots which are subjected to the impurities test of 4.5.6 shall be analyzed for aluminum only.

4.4 Examination. The samples selected in accordance with 4.3.2 shall be examined to verify conformance with all requirements of this specification which do not involve tests.

4.4.1 Material certification. Certification shall be provided as to conformance with requirements specified in 3.3 through 3.3.3. This certification shall document actual inspection or other verifiable quality data. Copies of the certification documentation shall be forwarded to the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782 and to the procuring activity.

4.5 Test procedures.

4.5.1 Shock tests.

4.5.1.1 Dome. A steel ball weighing 1/2 pound shall be dropped, free fall, from a height of 3 feet onto the center of the top of the dome.

4.5.1.2 Complete assembly. A completely assembled flash arrester shall be subjected to the grade A, class 1, type I shock test specified for lightweight equipment in accordance with MIL-S-901. For this test, the flash arrester shall be installed in a submarine cell jar cover which in turn shall be mounted on the anvil plate shown on figure 4 (entitled "Anvil plate of shock-testing machine for lightweight equipment") of MIL-S-901 through the terminal post holes in the cover. Other means of mounting the flash arrester to the shock test machine which duplicate this method may be employed.

4.5.2 Torque test. Twice a week throughout the life test of 4.5.3, the flash arrester shall be completely unscrewed from the cell cover in which it is installed, then it shall be reinstalled in the cover and tightened hand-tight.

4.5.3 Life test. Flash arresters on life test shall be installed on submarine main storage battery cells undergoing test at a laboratory satisfactory to NAVSEC. They shall remain installed on as many groups of cells as may be necessary to cause the flash arresters to be in service for a total of ampere-hours of cell charging and at currents not less than as specified on the applicable specification sheet. They shall be removed every six months, rinsed thoroughly with fresh water, dried for four hours in an oven at a temperature of 125° ± 5°F. and replaced immediately on their respective cells. Upon completion of the life test, flash arresters shall be subjected to the flash arrestment test (see 4.5.5).

4.5.4 Pressure drop test.

4.5.4.1 Dry test. Air shall be vented through a dry flash arrester at the rate specified on the applicable specification sheet. The pressure drop experienced by the air passing through the dome shall be measured with a water manometer.

4.5.4.2 Wet test. Submerge each submit flash arrester in sulphuric acid of 1.215 specific gravity for 2 minutes at an acid temperature of 80°F. No sooner than 15 seconds nor later than 60 seconds after being removed from the acid bath, during this time period the flash arrester shall be in the inverted position to permit drainage, air shall be vented through the flash arrester at the rate specified on the applicable specification sheet. The pressure drop experienced by the air in passing through the dome shall be measured with a water manometer.

4.5.5 Flash arrestment test. While a gas mixture consisting of 67 percent hydrogen and 33 percent oxygen is passed through the flash arrester from the under (concave) side at the rate specified on the applicable specification sheet, an electric spark shall be generated continuously within 1/2 inch of the outside (convex side) of the flash arrester for 15 seconds.

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4.5.6 Impurities test. A flash arrester dome, without hard rubber base, shall be refluxed for 5 hours in one liter of sulphuric acid of 1.265 specific gravity at an acid temperature of 150°F. At the end of the refluxing period, the dome shall be rinsed with distilled water until the rinse water exhibits a pH of 6.5 or higher. The rinse water shall all be collected and added to the refluxed acid. This solution shall then be reduced to a volume of one liter by evaporation and analyzed to determine the amounts of the materials listed in table I (see 3.9).

4.5.7 Hot acid test. A 50-gram sample of flash arrester dome material shall be placed in a 250 ml. beaker filled with 150 ml. of 1.265 H_2SO_4. The beaker should be covered and heated at 220°F. for 2 hours. The electrolyte level should be marked at the beginning of the test and maintained throughout by the addition of distilled water. Upon cooling, examine solution for presence of any silica (waxy, gelatinuous, white precipitate) which floats. Presence of any floating silica material shall be cause for rejection.

4.6 Where contracts or orders involve 50 or less flash arresters, quality conformance tests specified in 4.5.4 through 4.5.7 may be waived, provided a certification documenting actual inspection or other verifiable quality data is furnished. In no case shall examination specified in 4.4 be waived.

4.7 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation and packaging (see 6.1).

5.1.1 Level A. Flash arresters shall be protected and packaged in accordance with MIL-P-116, method III and as specified in 5.1.1.1 and 5.1.1.1.2.

5.1.1.1 Quantities less than 42. Each arrester shall be packaged in a box conforming to one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-566	
PPP-B-636	Domestic
PPP-B-665	
PPP-B-676	

Box closure shall be as specified in the applicable box specification or appendix thereto. Arresters shall be cushioned to prevent damage during handling, shipment, and storage. Alternatively, flash arresters individually cushioned may be packed in containers for the level of packing specified.

5.1.1.1.2 Quantities of 42 or more. A quantity of 42 flash arresters each individually cushioned and the entire quantity adequately blocked or braced as required to prevent damage from handling, shipment, and storage shall be packed in containers for the level of packing specified.

5.1.1.2 Level C. Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration or physical damage. This level may conform to the supplier's commercial practice.

5.1.2 Packing (see 6.1).

5.1.2.1 Level A. Flash arresters, packaged as specified, shall be packed in containers conforming to any one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-585	Class 3
PPP-B-591	Weather resistant
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-636	Weather resistant
PPP-B-640	Class 2

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Shipping containers shall have caseliners conforming to MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Caseliners for fiberboard boxes conforming to PPP-B-636, weather resistant class, may be omitted provided all corners and edge seams and manufacturer's joints are sealed with minimum 2 inch wide tape conforming to PPP-T-76. Boxes shall be closed, strapped or banded in accordance with the applicable box specification or appendix thereto.

5.1.2.2 Level B. Flash arresters, packaged as specified, shall be packed in containers conforming to any one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-585	Class 1 or 2
PPP-B-591	Domestic type
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-636	Domestic
PPP-B-640	Class 1

Box closures shall be as specified in the applicable box specification or appendix thereto.

5.1.3 Marking. In addition to any special marking, requirements, interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

5.2 Use of polystyrene (loose-fill) material.

5.2.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent flexible cellular material."

5.2.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification and applicable specification sheet.
(b) Type required (see 1.2 and 3.2).
(c) Quality assurance requirements (see 4.1.1).
(d) Preservation, packaging, packing or marking requirements (see 5.1.1).
(e) When polystyrene (loose-fill) material is approved (see 5.2.1).

6.2 Management control system document. The following management control system document should be included on DD Form 1660:

(a) MIL-I-45208 (see 4.1.1).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-22681 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattaville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

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6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 THE MARGINS OF THIS SPECIFICATION HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 6140-N329)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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